Interventions to reduce symptoms of common mental disorders and suicidal ideation in physicians: a systematic review and meta-analysis

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Summary

Background **An increased prevalence of common mental disorders and suicide has been reported among physicians worldwide. We aimed to assess which, if any, interventions are effective at reducing or preventing symptoms of common mental health disorders or suicidality in physicians.**

Methods **For this systematic review and meta-analysis MEDLINE, EMBASE, PsycINFO, and Cochrane CENTRAL (database inception to March 26, 2018), reference lists of included studies, and additional sources were systematically searched and screened by two independent reviewers. We included randomised controlled studies or controlled before–after studies of interventions to reduce depression, anxiety, or suicidality in physicians, as assessed by a validated outcome measure. Both organisation-level and physician-directed interventions were considered. Our primary outcome was differences in symptoms of common mental health disorders following intervention. We used random-effects modelling for the main meta-analyses and planned subgroup and sensitivity analyses. The study protocol is registered with PROSPERO, number CRD42018091646.**

Findings **We identified 2992 articles for screening, of which eight were included in the systematic review (n=1023 physicians) and seven in the meta-analysis. Results indicated a moderate effect in favour of the physician-directed interventions for reduction in symptoms of common mental health disorders (standardised mean difference 0·62; 95% CI 0·40–0·83; p<0·0001). Separate analyses showed physician-directed interventions resulted in reductions of symptoms of depression, anxiety, and suicidality. No evidence of significant heterogeneity was found (Q=3·78; p=0·44).**

Interpretation **Physician-directed interventions are associated with small reductions in symptoms of common mental health disorders among physicians. Research regarding organisational interventions aimed at improving physicians' mental health via modification of the work environment is urgently needed.**

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Introduction

Although being a physician is often described as one of the most rewarding professions, it is also known to be one of the most stressful and demanding occupations that, at times, can have a negative effect on the doctor's own mental health and wellbeing. Increasing evidence shows that physicians have increased prevalence of common mental disorders compared with the general population[1,2] and other workforces,[3] including symptoms of depression,[1] anxiety,[4] and suicidal ideation.[5] Further evidence of the burden of mental disorders among physicians comes from international evidence indicating that physicians are at greater risk of suicide than most other professional groups.[6–8] Although other occupational groups report more mental illness symptoms than average workers (eg, first responders[9,10]), none of these groups have a prevalence of death by suicide as high as physicians, with this difference becoming even greater once the prevalence has been adjusted for severity of mental illness, income, and other sociodemographic factors.[6] One physician is estimated to die by suicide each day in the USA.[11]

A range of workplace factors are likely to have a role in explaining the elevated prevalence of mental disorder and suicidality among physicians,[12] including a large workload, long and irregular working hours, competitiveness of training programmes, pressure of patient and service demands, the consequences of any errors, poor work–life balance, and the risk of moral injury if physicians are forced to work in ways that conflict with their ethics and values.[3,12] Broader systemic issues in medicine, such as stigma, regulatory practices, and concerns around confidentiality and registration, also affect the culture and workplace context physicians operate within and might act as barriers to help-seeking or disclosure through traditional pathways. These features make physicians a unique population and suggest that the type of

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Research in context

Evidence before this study

A high prevalence of common mental health disorders and suicide has been reported among physicians worldwide, compared with the general population and other workforces. This large mental health burden has negative effects on the individual and the health system more broadly. The mental health of physicians reflects a complex interaction between individual, workplace, and systematic factors within medicine, such as long working hours, excessive workload and administrative demands, regulatory practices around licensing and reporting, and stigma towards mental illness. Increasing concern from the international medical community on this issue has seen calls for action demanding a greater focus on physician wellbeing in medical training, workplaces, and the health system. However, which, if any, interventions are effective at reducing or preventing depression, anxiety, or suicidality among physicians remains unclear. We searched MEDLINE, EMBASE, PsycINFO, and Cochrane CENTRAL from database inception to March 26, 2018, for English-language publications using search terms related to three broad areas: "mental disorder/symptoms (including depression/anxiety)", "physicians/doctors", and "RCTs/interventions". Eight studies were identified of acceptable quality.

Added value of this study

To our knowledge, this systematic review and meta-analysis examines, for the first time, the effectiveness of controlled interventions in reducing symptoms of common mental health disorders and suicide among physicians. Our results indicate a concerningly small evidence base overall. We found encouraging findings for interventions addressed at the individual physician. Additionally, we identified an absence of controlled studies examining organisational-level interventions, such as modifying working hours.

Implications of all the available evidence

These findings should serve as a guide to the types of individual-focused interventions that could be provided as part of physicians' ongoing training. Our review highlights an urgent need for further research on organisational interventions aimed at reducing mental health symptoms in physicians through modification of the work environment. Given the mental health morbidity among physicians, more research is required to inform workplace practice and policy and to improve individual wellbeing in this population.

intervention effects seen in other occupational groups cannot be assumed to be present among physicians. Multifactorial causal models posit that employee mental health reflects the interaction of these workplace factors with other environmental and individual-level variables to shape outcomes for each physician.[13,14] Any attempt to improve the mental health of physicians will require interventions targeted not just at individuals but at an organisational or health systems level.[15,16]

Most of the research on physicians' mental health published to date has focused on broad proxy measures of wellbeing,[17] such as burnout[16,18] and perceived occupational stress.[19] Although evidence exists that both stress and burnout are related to depression, anxiety, and suicidality, they are distinct constructs and might be affected by different workplace factors.[20] Whether interventions that have been shown to reduce stress and burnout also reduce symptoms of mental disorder and suicidality, and whether any suggested effects hold in physician-only samples remains unclear. For example, a systematic review of randomised controlled trials (RCTs) aimed at reducing stress or burnout among health-care workers found that mental and physical relaxation training was the most common intervention tested, accounting for more than half of the published trials.[19] Although these low-level brief interventions appear to be effective in reducing perceived stress or burnout, meta-analyses of general population studies have now shown that such interventions are less effective in addressing symptoms of mental disorder, such as depression, than more structured cognitive or behavioural programmes.[21,22] Thus, evidence for a trickle-down effect, in which improvements in wellbeing might reduce the incidence of mental disorders like depression, is scarce. The available evidence from reviews of interventions for burnout or stress in physicians[16,23] cannot be used to determine which, if any, strategies are effective for reducing the burden of mental disorder and suicide among this group.

Awareness of the substantial rates of mental health disorder is increasing among physicians as well as the effects they are likely to be having on individuals, families, and the health service more broadly,[24–26] including quality of patient care and medical errors.[27,28] The international medical community has made numerous calls for action, demanding a greater focus on physicians' mental health within training programmes, workplaces, and health services.[24–26] An essential early step in developing a response to this issue is evaluation of the evidence base. In this Article, we examine the effectiveness of interventions in reducing symptoms of common mental disorders, specifically depression, anxiety, and suicidality (including suicidal ideation and attempted or completed suicide) among physicians.

Methods

Search strategy and selection criteria

For this systematic review and meta-analysis, we searched four bibliographic databases from inception to March 26, 2018: MEDLINE (from 1946), PsycINFO (from 1806), EMBASE (from 1947), and Cochrane CENTRAL.

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The search strategy combined three key sets of terms pertaining to doctors or physicians, RCTs or interventions, and mental disorders or symptoms. A wide variety of terms for each of these three concepts were entered as keywords, text words, titles, and subject headings as appropriate for each database. Full search terms for each database are outlined in the appendix. The search was limited to studies of humans and those published in English-language, peer-reviewed journals only. The reference lists of eligible studies and of relevant sources (appendix) were manually examined to identify any additional relevant studies.

Eligibility criteria

The population of interest comprised physicians of any specialty working in primary, secondary, or tertiary care settings, including interns, residents, registrars, fellows, and postgraduate trainees (or equivalent). Although we acknowledge that the precise meaning of terms used to refer to doctors varies, for consistency and clarity, we applied the term physician to mean a medical practitioner of any specialty at any stage of training or employment. We excluded medical students, retired physicians, or samples comprising solely other health-care professions (eg, nurses) or the general working population. We included studies based on a mixed sample of physicians and other health-care professionals only when physicians constituted at least 70% of the overall sample, in line with the criteria used by another review.[16] When multiple publications from the same study population were available, we included the most recent study.

Eligible interventions were any intervention intended to reduce symptoms or cases of common mental disorders in physicians, specifically anxiety, depression, or suicidality (including suicidal ideation and attempted and completed suicide). We considered both universal and non-universal interventions, meaning trials aimed at both prevention and treating established symptoms were eligible. Interventions could be delivered in any format (eg, face to face or online) and be directed towards an individual or a group of physicians. Interventions targeting workplace factors at the organisational-level were also eligible, such as adjustments in working hours or roster rescheduling. Such interventions were termed organisational, whereas interventions directed at the individual physicians were termed physician-focused.

Eligible studies were those that compared outcomes of interest between an intervention group and any type of control group, either an active (alternative intervention) or non-active (waitlist) control group. Eligible study designs included RCTs, non-RCTs, controlled before–after studies, and interrupted time-series.

Studies were eligible if they assessed at least one of the following mental health outcomes with a validated measure: depression, anxiety, a combination measure of depression and anxiety (sometimes termed general psychological distress), or suicidal ideation or behaviour.

The protocol was registered online on PROSPERO before commencement (number CRD42018091646). This systematic review and meta-analysis adheres to the PRISMA statement (appendix).

For **PROSPERO** see https://www.crd.york.ac.uk/PROSPERO

Data analysis

After the removal of duplicate records, two reviewers (KP and STEB) independently did a screening and selection process. Titles and abstracts were screened for relevance, then full-text records of relevant studies were examined to identify studies eligible for inclusion in the review. Discrepancies were resolved through discussion at each stage, and consensus was achieved with acceptable inter-rater reliability ($\kappa=0\cdot67$; $p<0\cdot01$). A third author (SBH) verified the eligibility of all included studies.

See **Online** for appendix

One author (KP) extracted data using a customised spreadsheet to record relevant quantitative and descriptive information for each included study. The Cochrane Collaboration's tool for assessing risk of bias was used to evaluate the risk of bias of included studies.[29] Each of the seven risk criteria was scored against a three-point rating scale, corresponding to a high, low, or unclear risk of bias. The risk of bias appraisal was done independently by two reviewers (KP and STEB), with good inter-rater agreement ($\kappa=77$; $p<0\cdot001$) and consensus achieved through discussion.

We analysed quantitative data for mental health outcomes using random-effects meta-analyses and planned subgroup and sensitivity analyses in the Comprehensive Meta-Analysis version 3 program. The standardised mean differences (SMDs) and associated 95% CI for continuous measures of symptoms of mental ill-health (depression, anxiety, a combined measure of general psychological distress, and for an overall outcome of symptoms of common mental health disorders) were pooled and forest plots created in Comprehensive Meta-Analysis. The mental health outcome measures of depression, anxiety, and general psychological distress symptoms were examined with continuous data to calculate pooled SMD (95% CI) for each. Our primary analyses included three planned a-priori meta-analyses to evaluate the effect of the intervention on common mental health disorders symptoms, prevalence of probable common mental health disorders caseness, and prevalence of suicidal ideation (reported as pooled risk ratios [RR] with 95% CIs). Because most studies were expected to report on symptoms, the primary outcome was symptoms of common mental health disorders (SMD; 95% CI) for the main and subgroup analyses. Positive effect sizes indicate superior effects of the intervention versus the control group. When outcome data were available for more than one follow-up timepoint, data for the shortest assessment point were entered.

Initially, depression symptoms, anxiety symptoms, and a combined measure of these symptoms termed general psychological distress, were examined separately. When

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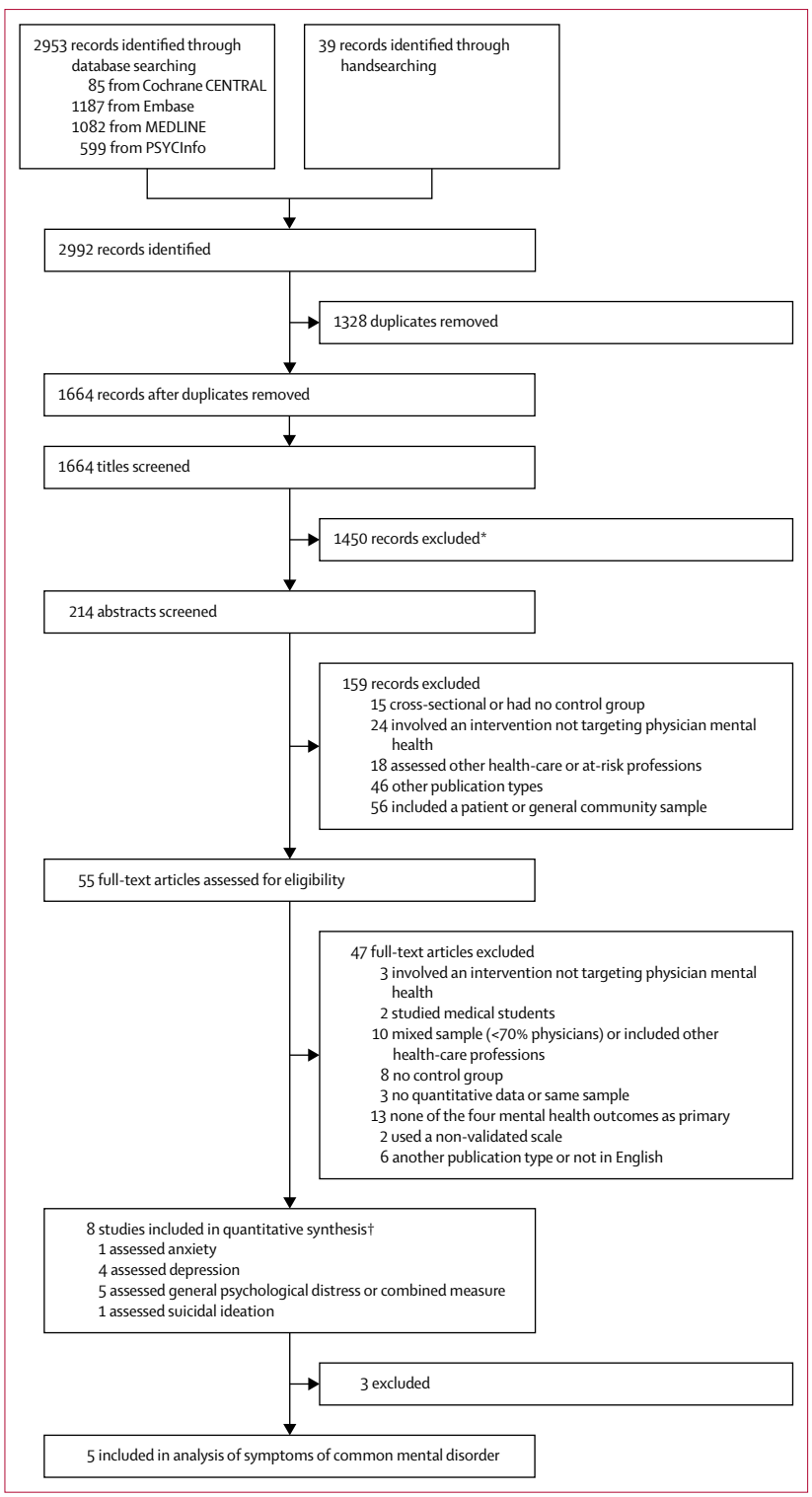

Figure 1: **Study selection**
*Including 35 reviews of interest for reference list screening. †Symptoms assessed are not mutually exclusive.

a single study reported multiple outcomes, specific continuous data with the appropriate outcome measure were entered and considered independently. Then, to maximise the number of studies used in the final and subgroup analyses, all measures of mental health symptomatology were combined into an overall outcome, termed symptoms of common mental disorders. When more than one of the prescribed measures was reported by a single study, priority was given to a combined measure of depression and anxiety, followed by a validated measure of depression, then anxiety. Suicidal ideation was considered as a separate outcome in a separate analysis of intervention effect.

All subgroup analyses were planned a priori and included comparisons of organisation compared with physician-focused interventions, comparison of the effectiveness of different types or modalities of interventions, and examination of differing effects among various physician specialties and different stages of training.

Two sensitivity analyses were done; the first to examine whether effects remained robust when only studies with low risk of bias ratings were retained in the analysis, and the second to examine the effect of removing the studies that used mixed samples and including only physician-only samples. Cohen's Q statistic of between-group variance was calculated, with a Cohen's Q statistic of $p<0\cdot01$ as the threshold for significant heterogeneity.[30] Publication bias was assessed using Egger's test for asymmetry, fail-safe number, and visual inspection of funnel plots.

Role of the funding source

None of the funders had any role in developing the study design, in the collection, analysis, and interpretation of data, in the writing of the report, or in the decision to submit the Article for publication. The corresponding author had full access to all the data and final responsibility for the integrity of the data, the accuracy of the data analysis, and the decision to submit for publication. The authors had total independence from the funders in their interpretation and reporting of the study results.

Results

The literature search yielded 2992 articles (figure 1). Following the removal of duplicates, 1664 articles were retained for title screening and 214 articles for abstract screening (figure 1). Of these articles, eight were identified as eligible and included in the review (figure 1; table).

The eight included studies contained data on a total of 1023 physicians (mean age range from 25·2 years [SD 8·1] to 47 [8·0] years) with relatively equal gender distribution (with the exception of one female-only sample; table). Participants comprised currently practising physicians across various specialties and stages of career, with two

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samples composed entirely of physicians in training. Five studies were done in the USA, two in Australia, and one in Spain (table). Seven studies were RCTs, and one was a controlled before–after study (table). Six of the interventions were delivered universally to any working physician, one was delivered only to working mothers,[36]

	Population	Setting and sample	Sample size at baseline; gender proportion (%); mean age in years, mean (SD)	Study design; type of prevention; control group	Intervention type;* delivery; approach†	Intervention	Assessment timepoints	Outcome of interest (scale); effectiveness‡
Asuero et al (2014);[31] Spain in 2010–12	Primary health-care professionals in clinical practice (physicians, nurses, social workers, and clinical psychologists)	Primary health-care centres; mixed sample (N=68): 41 (60%) physicians, 23 (33%) nurses, 4 (7%) social workers or clinical psychologists	68‡ (IG: n=43, CG: n=25); 5 men (8%) and 63 women (92%); 47 years (8·0)	RCT; universal; waitlist control	Physician-directed (group); group face-to-face delivery; CBT or mindfulness	8-week intervention: 8 weekly psychoeducation sessions (2·5 h per session); intensive 8-h guided silent mindfulness session (28 h in total)	Baseline, post-intervention (8 weeks)	Mood disturbance (POMS-short); total mood disturbance: effective (p<0·01), depression-dejection subscale of POMS: not effective
Dyrbye et al (2016);[32] USA in 2012	Physicians in practice at Department of Medicine or Surgery (specialty: internal medicine or surgery)	University-based hospital; physician-only (N=290)	290 (IG: n=145, CG: n=145); IG: 87 men (64%) and 58 women (36%), CG: 97 men (71%) and 48 women (29%); age reported as ranges	RCT; universal; no intervention	Physician-directed (individual); online; CBT or mindfulness	10-week online individualised intervention: weekly online menu of 5–6 self-directed micro-tasks, one task per week (5 mins per task; 0·5 h minimum)	Baseline, end of study (3 months)	Depression screen (PRIME-MD); positive depression screen: not effective
Gardiner et al (2004);[33] Australia; year not reported	GPs who elected to or had attended professional development courses (specialty: GP)	Metropolitan area, 1 Australian city; physician-only (N=100)	100 (IG: n=86, CG: n=24); gender not reported; age reported as ranges	CBA study; universal; no intervention (offered opportunity to participate in future stress management courses)	Physician-directed (group); group face-to-face delivery; CBT or mindfulness	5-week stress management training programme (15 h in total)	Pre-intervention, post-intervention (4 weeks), follow-up (12 weeks) of IG only	General psychological distress (GHQ-12); effective (p=0·001)
Guille et al (2015);[34] USA in 2009–10 or 2001–12 academic years	Medical interns entering internship year in traditional and primary care (speciality: various)	University-based hospitals; physician-only (N=199)	199 (IG: n=100, CG: n=99); 100 men (51%) and 99 women (49%); 25·2 years (8·1)	RCT; universal; attention-control (weekly email containing mental health information and contacts)	Physician-directed (individual); online; CBT or mindfulness	1-month intervention: 4 weekly psychoeducation-web online sessions (30 mins per session; 2 h in total)	Baseline (pre-internship and pre-intervention), 3, 6, 9, and 12 months	Suicidal ideation (item 9 of PHQ-9); positive endorsement of suicidal ideation: effective (p<0·05)
Holt and Del Mar (2006);[35] Australia in 1999	GPs with GHQ-12 score ≥3 indicative of current psychological distress (speciality: GP)	1 Australian state; across 8 Divisions of General Practice; physician-only (N=233)	233 (IG: n=120, CG: n=113); gender not reported; age not reported	RCT; indicated; no intervention	Physician-directed (individual); mailed; other	One-off mailed intervention in hardcopy; reminder letter 1 month later	Baseline, follow-up at 6 months (3 months post-intervention, 6 months post-baseline)	General psychological distress (GHQ-12); borderline significance (p=0·05)
Luthar et al (2017);[36] USA in 2015	Working female health-care practitioners (physicians, PhD clinicians, physician assistants, and nurse practitioners) who were mothers with at least one child <18 years of age (specialty: various)	University-based clinic; mixed sample (N=40); 26 (65%) physicians or PhD clinicians, 14 (35%) physician assistants or nurse practitioners	40§ (IG: n=21, CG: n=19); no men and 40 women (100%); IG: 38·8 years (6·1), CG: 39·4 years (4·8)	RCT; selective; no intervention (12 h in total; 1 h per week protected free time)	Physician-directed (group); group face-to-face delivery; other	3-month intervention: 12 weekly facilitated group sessions (1 h per session; 12 h in total)	Baseline, post-intervention, follow-up (3 months) for psychological measures plus plasma cortisol	Depression (BDI) and general psychological distress (BSI); depression: effective (p <0·05), global symptoms of distress: effective (p<0·05)

(Table continues on next page)

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	Population	Setting and sample	Sample size at baseline; gender proportion (%); mean age in years, mean (SD)	Study design; type of prevention; control group	Intervention type;* delivery; approach†	Intervention	Assessment timepoints	Outcome of interest (scale); effectiveness‡
(Continued from previous page)								
Saadat et al (2012);[37] USA in 2006–08 academic years	Anaesthesiology residents (first to third year; speciality: anaesthesiology)	University-based teaching hospital; physician-only (N=60)	60 (WIG: 19 [33%], NTC-RT: 20 [33%], NTC-RD: 19 (33%); WIG: 9 men (47%) and 10 women (53%), NTC-RT: 11 men (55%) and 9 women (45%), NTC-RD: 11 men (58%) and 8 women (42%); WIG: 30·7 years (2·3), NTC-RT: 32·1 years (4·8), NTC-RD: 31·1 years (2·6)	RCT; universal; attention-control (NTC-RT): relieved from clinical duties, gathered in single room to spend 16 consecutive 1·5 h weekly sessions as free time; no intervention (NTC-RD): routine duties	Physician-directed (group); group face-to-face delivery; CBT or mindfulness	16-week course: 16 weekly sessions (1·5 h per session; 24 h in total)	Pre-intervention, post-intervention (16 weeks)	Depression (CES-D) & anxiety (STAI-S); depression: not effective, anxiety: effective (p <0·05)
West et al (2014);[38] USA in 2010–2012	Physicians in practice at Department of Medicine (speciality: internal medicine)	University-based clinic; physician-only (N=74)	74 (IG: n=37, CG: n=37); IG: 25 men (68%) and 12 women (32%), CG 24 men (65%) and 13 women (35%); age not reported	RCT; universal; attention-control: 1 h protected free time	Physician-directed (group); group face-to-face delivery; other	9-month curriculum of 19 fortnightly 1-h sessions (19 h in total)	Baseline, every 3 months during the study, post-intervention follow-up (3 and 12 months)	Depression screen (PRIME-MD); positive depression screen: not effective

IG=intervention group. CG=control group. RCT=randomised controlled trial. CBT=cognitive behavioural therapy. POMS-short=Profile of Mood States Questionnaire-short (15-item). PRIME-MD=Primary Care Evaluation of Mental Disorders (9-item) depression screener questionnaire. GP=general practitioner. CBA=controlled before–after study. GHQ-12=General Health Questionnaire (12 items). PHQ-9=Patient Health Questionnaire. BDI=Beck Depression Inventory (21-items). BSI=Brief Symptoms Inventory. WIG=wellness intervention group (intervention group). NTC-RT=no-treatment control group with release time (control group 1). NTC-RD=no-treatment control group with routine duties (control group 2). CES-D=Center for Epidemiologic Studies Depression Scale. STAI-S=state anxiety subscale of Spielberger State-Trait Anxiety Inventory. *Physician-directed (group) refers to an intervention delivered to physicians on a group basis, whereas Physician-directed (individual) refers to an intervention delivered to an individual physician. †Approach indicates the main theoretical basis for intervention content reported by each study. Categories (ie, CBT or mindfulness or other) indicate classification of approach used in subgroup analyses of this variable. The CBT or mindfulness category could include interventions based on either CBT or on mindfulness. ‡Effectiveness was determined for each outcome measure of interest and denoted as being effective of not effective. When a study reported a significant difference in a particular outcome between intervention and control groups comparing preintervention to postintervention, it was classified as effective. Non-significant pre–post comparison results were classified as not effective. §Overall sample of all health professionals.

Table: **Characteristics of included studies**

and another only to physicians who scored above three on the General Health Questionnaire-12 (table).[35] Although no clear treatment studies were identified, none of the studies specifically excluded those with existing high severity of symptoms, meaning the samples for each study were composed of a mix of symptomatic and non-symptomatic individuals.

Five interventions were delivered through a group-based format and three on an individual basis (table). The content and theoretical approach of each intervention varied considerably (table). For simplicity and subgroup analyses, we classified the physician-focused interventions into two groups: interventions based on cognitive behavioural therapy (CBT; n=2) or mindfulness (n=2) principles and interventions based on other approaches (eg, a relational supportive and a coping mechanisms approach; n=3; table). Our search did not identify any controlled trials of organisational-level interventions.

The results of the risk of bias assessment are presented in the appendix. Four studies (50%) were assessed as low risk of bias, three as high risk of bias, and one as unclear (appendix). Studies most commonly did not clearly describe blinding procedures; however, overall, studies and evidence they reported were of acceptable quality (appendix).

As a first step, we examined each of the three mental health measures as separate outcomes, calculating pooled intervention effects for symptoms of depression, anxiety, or general psychological distress (figure 2; appendix). Each of the three measures had moderately sized positive intervention effects (figure 2), with the caveat of being based on only a small number of studies. Specifically, results showed a reduction in symptoms of depression for the intervention groups compared with controls (n=3; SMD 0·53; 95% CI 0·20–0·87; p=0·0019), in general psychological distress (n=4; 0·65; 0·38–0·91; p<0·0001), and in anxiety, although this was based on one study (n=1; 0·71; 0·05–1·36; p=0·035; figure 2).

Five studies examined the effect of an intervention on severity of symptoms of common mental health disorders. For studies assessing symptoms of depression, anxiety, or a combined measure, the overall SMD between the intervention and control groups was 0·62 (95% CI 0·40–0·83; p<0·0001; figure 3; appendix),

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indicating a moderate effect in favour of the intervention. We found no evidence of significant heterogeneity (Q=3·78 [4]; p=0·44). When only studies with low risk of bias were retained in the analyses (n=4), the intervention effect was almost identical (SMD 0·65; 95% CI 0·19–1·10). When only studies using physician-only samples (n=3) were retained in the analysis, the intervention effect for symptoms of common mental disorder remained of similar size (SMD 0·56; 95% CI 0·25–0·86; p<0·0001).

Although several subgroup analyses were planned a priori, only three subgroup analyses could be done given the small number of eligible studies identified, and even within these subgroup analyses, formal testing of differences was not always possible.

The pooled mean effect size for the group-based interventions (n=4; SMD=0·78; 95% CI 0·50–1·06) was larger than that obtained in the main analyses and when compared with the one individual-based study (n=1; 0·39; 0·05–0·72), although with overlapping 95% CIs.

The pooled mean effect size for studies with a non-active control group (either no intervention or waitlist control) was 0·62 (n=4; 95% CI 0·36–0·88) compared with the single active control group design (consisting of allocated free time from clinical duties in a single room; n=1; SMD 0·74; 0·10–1·39), with no between-group difference identified (Q=0·13 [1]; p=0·72).

The effect size for using CBT or mindfulness-based intervention content was 0·79 (n=3; 95% CI 0·47–1·11), whereas interventions based on other approaches yielded an effect size of 0·46 (n=2; 0·16–0·76); however, the between-group difference was not significant (Q=2·14 [1]; p=0·14).

Two studies assessed a dichotomous outcome measure of depression with the two-item Primary Care Evaluation of Mental Disorders depression screener questionnaire scale[38] to indicate probable depression, by use of a mindfulness-based approach[18] or a broad-based curriculum.[32] A pooled relative risk estimate of 0·81 (n=2; 95% CI 0·57–1·16; p=0·25; Q=52; p>0·05) was obtained.

Only one study[34] assessed suicidal ideation, finding that intern physicians randomly assigned to a web-based CBT intervention group were 60% less likely to report suicidal ideation during an internship year than the attention-control group (RR=0·40; 95% CI 0·17–0·91; p=0·03).

Visual inspection of the funnel plot (appendix) did not indicate any evidence of asymmetry, with a fail-safe number of 38 studies and a non-significant Egger's test of the intercept (t=1·6; p=0·2) confirmed the absence of significant publication bias.

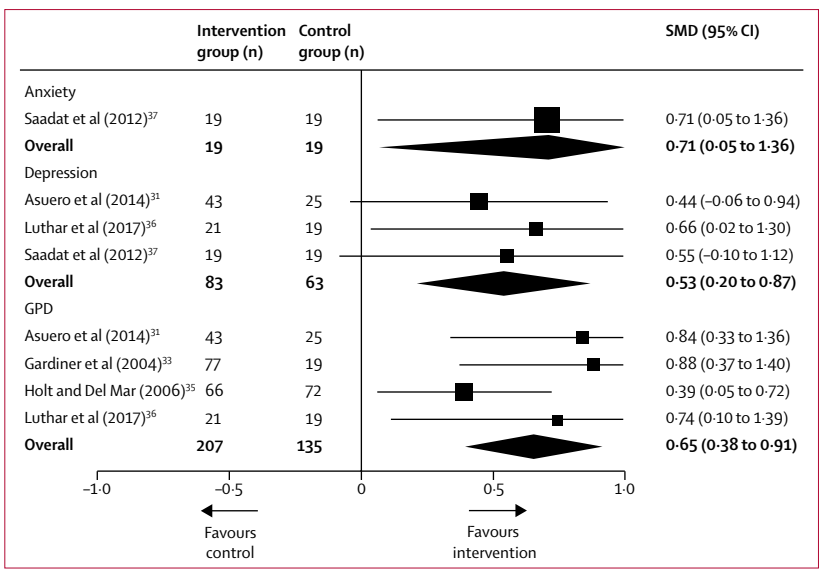

Figure 2: **Forest plot of the effects of interventions by each mental health outcome measure in physicians**
SMD=standardised mean difference. GPD=general psychological distress.

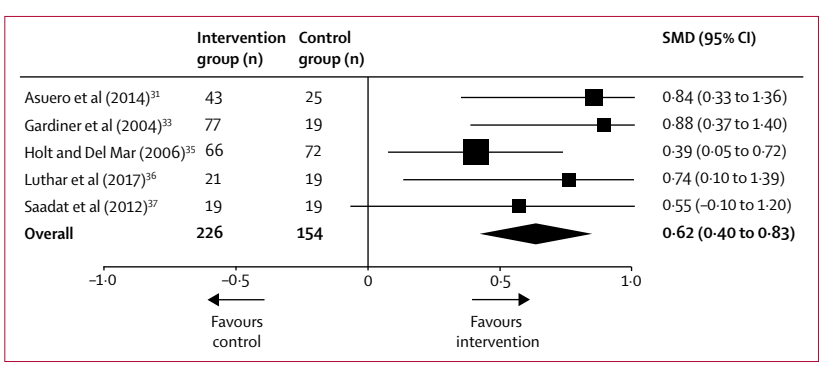

Figure 3: **Forest plot of the effects of interventions on symptoms of common mental helath disorders in physicians**
SMD=standardised mean difference.

Discussion

To our knowledge, this Article represents the first systematic review and meta-analysis to evaluate the effectiveness of interventions designed to reduce symptoms and prevalence of common mental disorders and suicidal behaviour among physicians. This review has two key findings. First, remarkably little published research examines the efficacy of potential solutions. We were not able to identify any controlled studies of organisation-level interventions and only a few well-conducted trials of programmes directed towards physicians, either on an individual or group basis. Second, the available research data suggest that interventions directed towards physicians can be effective in reducing mental health symptoms, when considered as a combined outcome representing common mental disorders or separately as individual measures. Our main analysis showed that individual-level interventions reduced symptoms of common mental disorders with a moderate effect. We found evidence for positive intervention effects for symptoms of depression, general psychological distress, anxiety, and suicidal ideation, although outcomes for anxiety and suicidal ideation were based on single studies. The effect sizes observed for individual-level interventions were a similar moderate size in favour of the intervention

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across each of these different outcomes (effect sizes range 0·53–0·71). It must be noted that each of these outcomes is dependent on self-reporting of symptoms. We were able to identify only two studies that considered caseness of common mental disorders, rather than just the overall symptoms, and we found no studies that used clinician diagnosis. The fact that pooled effect size from studies with caseness as an outcome were not statistically significant raises the possibility that individual-level interventions are able to reduce average symptom severity, but might not be able to prevent new cases of depression or anxiety. Similarly, no clear treatment or pure prevention trials were reported, meaning that it is not known whether the symptom reduction observed is the same across the spectrum of symptoms. A disconnect can occur between symptoms and functional improvement,[39] meaning that a reduction in symptoms might not necessarily lead to improved functioning and reduced sickness absence.

The absence of any controlled research studies examining organisation-level interventions is of particular concern. Organisational-level interventions have the advantages of directly addressing workplace risk factors, the potential to be more acceptable to participants, and acting as a natural home for universal prevention.[40] The importance of providing both individual and organisational or structural solutions is now well-established for creating more mentally healthy workplaces.[41,42] Among other high-risk occupational groups (eg, first responders) a range of evidence-based interventions are aimed at the broader organisation, rather than the individual employees only.[42,43] The potential for organisational-level interventions in a health-care context has been highlighted by reviews of controlled interventions to reduce burnout among physicians[16,18] and health-care workers.[19] Their findings suggest that organisational approaches, such as re-scheduling of work hours, reducing workloads, and modifying local working conditions, can lead to modest reductions in burnout and work-related stress. These types of organisational-level interventions could also prevent future problems rather than simply being a reaction to emerging distress. Institutional approaches that improve the values of the work environment and draw on available evidence could inform the design of future interventions for physicians.

Despite the paucity of research on how mental health of physicians can be enhanced, previous reports have highlighted that physicians are by far the most studied of all health professionals.[44] Thus, even less is likely to be known about how to address the mental health of nurses, psychologists, and other allied health professionals. The lessons learnt from studies examining physicians' mental health can hopefully be of use to other health professional groups as well.

Our positive findings regarding individual-level interventions raise the need for consideration of how these evidence-based approaches can best be integrated into the ongoing training and education of physicians. Co-design and user experience strategies are used increasingly by researchers to develop interventions tailored to the specific needs of a target population. These strategies enhance acceptability, feasibility, and participant engagement, all of which are essential to intervention efficacy and adherence. They should inform how best to provide evidence-based individual-level interventions and guide the development of new organisational-level approaches to modification of some of the unique workplace factors proposed as potential reasons for physician mental ill-health, such as work hours, workplace culture, regulatory requirements, and the conflicting demands of administrative tasks, ongoing training, and delivery of care to patients. These approaches also need to take into account the cultural and regulatory climate of the medical system as a core contributor to poor mental health of physicians. The threat of losing one's medical licence or mandatory reporting intensify an environment of shame, stigma, low amounts of help-seeking, and reluctance or fear regarding disclosure of mental health whether current or previous.[45] Organisational-level interventions also need to consider how recent trends in health care, such as an increasing administrative burden, might have affected the values, work–life balance, spirit, and dignity of physicians.[46] This wider context must be addressed in tandem with modifiable workplace factors and physician-directed strategies to respect the complex environment within which physicians operate, and to acknowledge their individual values and sense of meaning derived from their role, while avoiding placing any implicit or explicit blame on individual physicians who become unwell.

Teaching CBT or mindfulness techniques usually requires ongoing training and practice, which are both time consuming, expensive, and logistically challenging to deliver. Given the increasing evidence for the effectiveness of eHealth interventions, both for treatment and prevention of mental health problems,[47,48] the use of online or smartphone technology to deliver interventions among physicians needs to be examined. Additionally, online technology might address concerns reported by physicians around confidentiality, privacy, and stigma and mitigate small sample sizes and low participation by improving the reach and accessibility of the intervention.

This systematic review has several limitations. First, only eight eligible studies were identified, resulting in reduced power particularly for subgroup comparisons. Second, the studies were very diverse, differing in intervention approach, delivery, and the outcomes assessed. A methodological limitation of the individual studies was the short-term follow-up, with few studies assessing physicians' mental health beyond 3–6 months after the intervention. Third, although we chose to focus on only validated measures of mental health, all the outcome measures were based on self-report symptom scales rather than a clinical assessment. Fourth, the samples assessed are unlikely to be representative of all physicians, given the potential for self-selection bias, relatively small

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sample sizes, and the fact that most studies were done in only one location or setting. Fifth, although a second independent researcher undertook study selection and the risk of bias assessment, only one person extracted the data from each study, creating a potential for error or bias. Finally, we included English language-only publications, meaning studies published in other languages would not have been captured by this review.

Awareness of the substantial rates of mental health disorder among physicians as well as the effects they are likely to be having on individuals, families, and the health service more broadly,[24–26] including quality of patient care and medical errors,[27,28] is increasing. Although the scarcity of controlled studies in this area is concerning, particularly with regard to interventions at an organisational level, the available research data suggest that some interventions directed at physicians can reduce mental health symptoms. Given the prevalence of mental health morbidity among physicians, these findings should both guide the type of physician-focused interventions that are adopted among this group and serve as a call to action for the urgent need for more comprehensive rigorous research regarding individual and organisational interventions aimed at improving the mental health of physicians.

Contributors

KP and SBH designed the study after HC and SBH secured funding. KD and SBH planned the statistical analysis. KP analysed and extracted data, with assistance from KD and SBH. KP and STEB assessed study eligibility and did quality assessments. SBH monitored the review process. KP and SBH wrote the first draft of the manuscript. All authors contributed to the interpretation and subsequent edits of the manuscript. SBH is the guarantor.

Declaration of interests

All authors declare support from the Australian Government, iCare Foundation, and NSW Health for the submitted work. KP, JC, STEB, HC, and SBH are employed by the Black Dog Institute, a not-for-profit research institute that provides mental health training to a range of organisations. JaR was previously the chair of Doctors Health Service which is funded by the Medical Board.

Data sharing

Extracted data are available on request to the corresponding author.

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